# Theoretical Models of Hippocampal Information Encoding

## Overview

**Cell categories.** In the independent and place-dependent models, every neuron $i$ falls into exactly one of four categories, determined by two independent boolean draws: whether it carries a place field (probability `percent_place_cells`) and whether it is TEBC-responsive (probability from `percent_task_responsive_cells`):

| Category | Place field? | TEBC-responsive? |
|---|---|---|
| place-only | yes | no |
| place+task | yes | yes |
| task-only | no | yes |
| silent | no | no |

Only **place+task** cells blend a task and a place signal; place-only cells fire purely off the place signal, task-only cells fire purely off the task signal, and silent cells sit at a fixed near-zero baseline. The arousal-mediated model has no such categories: every cell is a place cell, and its task response (if TEBC-responsive) is selected by the animal's instantaneous running/in-field state instead (see §5).

**Balance parameter $\beta_i$.** Each *place+task* neuron $i$ has a balance parameter $\beta_i \in [0, 1]$ drawn from a population distribution $P_{\text{balance}}$, controlling the relative contribution of the task-driven signal versus the place-cell signal. $\beta_i = 1$ means the neuron's task signal fully dominates; $\beta_i = 0$ means it behaves as a pure place cell. $\beta_i$ has no effect on place-only, task-only, or silent cells.

**Model summary (place+task category only).**

| Model | High-level equation | Description |
|---|---|---|
| Independent | $r_i = \beta_i\, r_i^{\text{task}}(t) + (1-\beta_i)\tilde{r}_i^{\text{place}} + \varepsilon_i$ | The task signal is independent of the animal's current location and determined strictly as a function of the time since the onset of the stimuli. Each task-influenced neuron is randomly assigned 1 of 8 response functions, always scaled to a fixed peak amplitude. |
| Place-Dependent | $r_i = \beta_i\, r_i^{\text{task}}(t, \tilde{r}_i^{\text{place}}) + (1-\beta_i)\tilde{r}_i^{\text{place}} + \varepsilon_i$ | The task signal is dependent on the animal's current location: it is scaled to the neuron's own live place rate rather than a fixed amplitude. Each task-influenced neuron is randomly assigned 1 of 8 response functions. |
| Arousal-Mediated | $r_i = \tilde{r}_i^{\text{place}} + r_i^{\text{task}}(t, \tilde{r}_i^{\text{place}}) + \varepsilon_i$ | The task response is selected from one of 4 functions based on whether the rat is running and whether the cell is in a place field, and is additive on top of the place rate rather than blended with it. |

**Key symbols.**

- $\tilde{r}_i^{\text{place}}$ — velocity-modulated Gaussian place cell rate
- $r_i^{\text{task}}(t)$ — TEBC task response for neuron $i$, drawn from type $k \in \{1, \ldots, 8\}$ (independent/place-dependent) or $k \in \{1, \ldots, 4\}$ (arousal-mediated)
- $A_i$ — per-neuron task response *peak amplitude*: the ceiling the normalized response shape (see §4) is rescaled to. Model-specific; see §2/§3.
- $\beta_i$ — per-neuron balance parameter (place+task cells only)
- $b_i \equiv \tilde{r}_i^{\text{place}}(t)$ — live place rate, used as the task amplitude $A_i$ in the place-dependent and arousal-mediated models
- $\varepsilon_i \sim \mathcal{N}\big(-\frac{0.02}{30}, \frac{0.02}{30}\big)$ — additive noise term, drawn independently per neuron per timestep, common to all models

**Notation.** All firing rates are in Hz. Time $t$ is seconds since CS onset. Constants: $t_{CS} = 0.25\,\text{s}$, $t_{US} = 0.75\,\text{s}$, $d_{US} = 0.1\,\text{s}$.

## Place Cell Firing Rate (RatInABox)

All models inherit the Gaussian place cell implementation from the RatInABox library. For each neuron $i$ with field centre $\mathbf{c}_i$ and width $\sigma$:

$$r_i^{\text{PC}}(\mathbf{x}) = \exp\left(-\frac{d(\mathbf{x},\, \mathbf{c}_i)^2}{2\sigma^2}\right) \cdot (f_{\max} - f_{\min}) + f_{\min} \tag{1}$$

where $d(\mathbf{x}, \mathbf{c}_i)$ is the geodesic distance from the agent position $\mathbf{x}$ to the place field centre and $\sigma = 0.20\,\text{m}$. $f_{\min} = 0\,\text{Hz}$ and $f_{\max} = 12\,\text{Hz}$ for *all three* models — the independent and place-dependent models share the same underlying place-cell construction (`CombinedPlaceTebc`), so independent no longer uses a reduced $f_{\max} = 1\,\text{Hz}$ ceiling.

## Independent Model

*The task signal is independent of the animal's current location and determined strictly as a function of the time since the onset of the stimuli. Each task-influenced neuron is randomly assigned 1 of 8 response functions, and the task amplitude is a fixed constant shared by every task-responsive neuron.*

**place-only cells:**

$$r_i(t) = \tilde{r}_i^{\text{place}}(t) + \varepsilon_i \tag{2}$$

**place+task cells** (in trial; falls back to the place-only equation outside a trial):

$$r_i(t) = \beta_i\, r_i^{\text{task}}(t) + \big(1 - \beta_i\big)\, \tilde{r}_i^{\text{place}}(t) + \varepsilon_i \tag{3}$$

**task-only cells** (in trial; falls back to baseline outside a trial):

$$r_i(t) = r_i^{\text{task}}(t) + \varepsilon_i \tag{4}$$

**silent cells:**

$$r_i(t) = \tfrac{0.02}{30} + \varepsilon_i \tag{5}$$

In all cases $\varepsilon_i \sim \mathcal{N}\big(-\frac{0.02}{30}, \frac{0.02}{30}\big)$ drawn independently per neuron, and $\beta_i \in [0, 1]$ is drawn from $\{\beta_i\} \sim P_{\text{balance}}$.

**Velocity-modulated place cell rate.** Let $r_i^{\text{PC}}$ be the standard Gaussian place cell firing rate and $v$ the agent's smoothed speed (m/s). The modulated rate is:

$$\tilde{r}_i^{\text{place}} = \begin{cases} \frac{0.02}{30} & v < 0.02 \\ \dfrac{p(100v)}{30}\, r_i^{\text{PC}} & v \geq 0.02 \end{cases} \tag{6}$$

where $p$ is the empirical cubic polynomial

$$p(s) = -3.261 \times 10^{-4}\, s^3 + 1.741 \times 10^{-2}\, s^2 + 8.366 \times 10^{-2}\, s + 1.161 \tag{7}$$

(argument $s = 100v$ converts m/s to cm/s). Non-place-cell neurons have their rate fixed at $\frac{0.02}{30}$ regardless of $v$.

**Task firing rate.**

$$r_i^{\text{task}}(t) = \begin{cases} R_{c_i}(t;\, A_i) & \text{neuron } i \text{ is TEBC-responsive} \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

where $c_i \in \{1, \ldots, 8\}$ is the cell type assigned to neuron $i$ and $A_i$ is the peak amplitude every response profile is rescaled to (§4). In the independent model $A_i$ **is the same fixed constant for every task-responsive neuron** (place+task and task-only alike): $A_i = 12\,\text{Hz}$, the shared population $f_{\max}$ (§1) — i.e. the response is not scaled by the individual cell's own place rate. The response profiles $R_k(t; A)$ map to the equations in §4 as follows:

| $k$ | Description | Equation |
|---|---|---|
| 1 | Bimodal Gaussian | (17) |
| 2 | One-sided Gaussian | (18) |
| 3 | Piecewise Linear Decay | (19) |
| 4 | Ramp with Gaussian Burst | (21) |
| 5 | Trapezoidal US Response | (22) |
| 6 | Noisy Uniform | (23) |
| 7 | Windowed Bimodal Gaussian | (25) |
| 8 | Post-CS Piecewise Decay | (26) |

## Place-Dependent Model

*The task signal is dependent on the animal's current location: rather than a fixed constant, its peak amplitude is the neuron's own live place rate. Each task-influenced neuron is randomly assigned 1 of 8 response functions.*

**place-only cells:**

$$r_i(t) = \tilde{r}_i^{\text{place}}(t) + \varepsilon_i \tag{9}$$

**place+task cells** (in trial; falls back to the place-only equation outside a trial):

$$r_i(t) = \beta_i \, r_i^{\text{task}}\big(t,\, \tilde{r}_i^{\text{place}}(t)\big) + \big(1-\beta_i\big)\, \tilde{r}_i^{\text{place}}(t) + \varepsilon_i \tag{10}$$

**task-only cells** (in trial; falls back to baseline outside a trial):

$$r_i(t) = r_i^{\text{task}}(t) + \varepsilon_i \tag{11}$$

**silent cells:**

$$r_i(t) = \tfrac{0.02}{30} + \varepsilon_i \tag{12}$$

$\varepsilon_i$ and $\beta_i$ are as defined in §2. Let $b_i \equiv \tilde{r}_i^{\text{place}}(t)$ be the neuron's own live velocity-modulated place rate:

$$r_i^{\text{task}}(t) = \begin{cases} R_{c_i}\big(t;\ A_i\big) & \text{neuron } i \text{ is TEBC-responsive} \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

$$A_i = \begin{cases} b_i & \text{place+task cell (has a place field)} \\ 0.02/30 & \text{task-only cell (no place field)} \end{cases} \tag{14}$$

i.e. a place+task neuron's task response peaks at its own live place rate; a task-only neuron (no place field to scale off) instead uses a small fixed placeholder amplitude. Each profile $R_k(t; A)$ has the same functional form as in the independent model (Equations (17)–(26)); the shape-construction magnitude parameters below are expressed as multiples of a unit baseline, matching the ratios the shape is built with before renormalization (§4):

| $k$ | Description | Shape magnitude ratios | Eq. |
|---|---|---|---|
| 1 | Bimodal Gaussian | $A_1 = 2.14, \quad A_2 = 5.71$ | (17) |
| 2 | One-sided Gaussian | $A = 3.14$ | (18) |
| 3 | Piecewise Linear Decay | $d_1 = 0.33, \quad d_2 = 0.17$ | (19) |
| 4 | Ramp with Gaussian Burst | $A_{CS} = 0.40, \quad A_{US} = 1.10$ | (21) |
| 5 | Trapezoidal US Response | $A_{US} = 3.14$ | (22) |
| 6 | Noisy Uniform | (unchanged — not renormalized) | (23) |
| 7 | Windowed Bimodal Gaussian | $A_1 = 1.67, \quad A_2 = 2.14$ | (25) |
| 8 | Post-CS Piecewise Decay | steps at 0.46, 0.62, 0.38 | (26) |

**Velocity modulation.** Identical to the independent model (same cubic polynomial $p(s)$ and the same $\frac{0.02}{30}$ floor below the speed threshold):

$$\tilde{r}_i^{\text{place}} = \begin{cases} \frac{0.02}{30} & v < 0.02 \\ \dfrac{p(100v)}{30}\, r_i^{\text{PC}} & v \geq 0.02 \end{cases} \tag{15}$$

## TEBC Response Profiles (Types 1–8)

The following profiles are shared by the independent and place-dependent models. Each type is built as a raw shape $S_k(t)$ using the magnitude constants given below (fitted assuming a unit baseline/amplitude of 1), then **renormalized** to run from 0 up to the neuron's actual peak amplitude $A_i$ (§2/§3):

$$R_k(t;A) = \frac{S_k(t) - \min_\tau S_k(\tau)}{\max_\tau S_k(\tau) - \min_\tau S_k(\tau)} \cdot A, \qquad \tau \in [-0.1,\, 2.0]\,\text{s (1000-point grid)} \tag{16}$$

**Type 6 is the one exception** — it is *not* renormalized; see (23). The raw shapes $S_k(t)$ below use the magnitude constants exactly as coded (independent model's fixed defaults; place-dependent's ratios from the table in §3), but only their *relative* shape survives (16) — the absolute constants shown do not appear directly in the final firing rate.

### Type 1 — Bimodal Gaussian

$$S_1(t) = A_1 \exp\!\left(-\frac{(t-\mu_1)^2}{2\sigma_1^2}\right) + A_2 \exp\!\left(-\frac{(t-\mu_2)^2}{2\sigma_2^2}\right) \tag{17}$$

Independent model constants: $A_1 = 0.15$, $\mu_1 = 0.3$, $\sigma_1 = 0.1$; $A_2 = 0.40$, $\mu_2 = 0.85$, $\sigma_2 = 0.1$; baseline $= 0.07$ (added into both Gaussian terms before normalizing). Place-dependent model uses $A_1 = 2.14$, $A_2 = 5.71$, same $\mu, \sigma$, baseline $= 1.0$.

### Type 2 — One-sided Gaussian

$$S_2(t) = \begin{cases} A \exp\!\left(-\dfrac{(t-\mu)^2}{2\sigma^2}\right) + b & t > \mu - 0.1 \\ b & \text{otherwise} \end{cases} \tag{18}$$

Independent: $A = 0.22$, $\mu = 0.75$, $\sigma = 0.15$, $b = 0.07$. Place-dependent: $A = 3.14$, same $\mu, \sigma$, $b = 1.0$.

### Type 3 — Piecewise Linear Decay

Independent: $b = 0.06$, $d_1 = 0.02$, $d_2 = 0.01$. Place-dependent: $b = 1.0$, $d_1 = 0.33$, $d_2 = 0.17$.

$$S_3(t) = \begin{cases} b & t \le 0 \\ \dfrac{d_1 - b}{0.2}\, t + b & 0 < t < 0.2 \\ \dfrac{d_2 - d_1}{0.65}(t - 0.2) + d_1 & 0.2 \le t < 0.85 \\ d_2 & 0.85 \le t \le 3.85 \\ b & t > 3.85 \end{cases} \tag{19}$$

### Type 4 — Linear Ramp with Gaussian Burst

Independent: $b = 0.075$, $A_{CS} = 0.03$, $A_{US} = 0.08$, $\sigma_{US} = 0.2$. Place-dependent: $b = 1.0$, $A_{CS} = 0.40$, $A_{US} = 1.10$, $\sigma_{US} = 0.2$. Define the unclamped response:

$$\tilde{S}_4(t) = \begin{cases} b & t \le 0.05 \\ \dfrac{A_{CS} - b}{0.73}(t - 0.05) + b & 0.05 < t \le 0.78 \\ A_{US} \exp\!\left(-\dfrac{(t-0.78)^2}{5\,\sigma_{US}^5}\right) + b & 0.78 < t \le 1 \\ A_{CS} & 1 < t \le 3 \\ b & t > 3 \end{cases} \tag{20}$$

The output is clamped to $[A_{CS},\, A_{US}]$ before normalizing:

$$S_4(t) = \min\big(A_{US},\ \max\big(A_{CS},\ \tilde{S}_4(t)\big)\big) \tag{21}$$

## Type 5 — Trapezoidal US Response

$$S_5(t) = \begin{cases} b & t < 0.75 \\ A_{US} & 0.75 \le t \le 0.78 \\ A_{US} + \dfrac{b - A_{US}}{0.42}(t - 0.78) & 0.78 < t < 1.2 \\ b & t \ge 1.2 \end{cases} \tag{22}$$

Independent: $b = 0.035$, $A_{US} = 0.11$. Place-dependent: $b = 1.0$, $A_{US} = 3.14$.

## Type 6 — Noisy Uniform (not renormalized)

$$R_6(t; A) = 0.5\, A + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(-\eta,\ \eta)\,, \quad \eta = \frac{0.03}{30} \tag{23}$$

Unlike types 1–5 and 7–8, this profile is applied directly to the peak amplitude $A$ (not passed through (16)).

## Type 7 — Windowed Bimodal Gaussian

Each Gaussian peak is truncated to a one-sided window:

$$G_i(t) = \begin{cases} A_i \exp\!\left(-\dfrac{(t - \mu_i)^2}{2\sigma_i^2}\right) & \mu_i - 3\sigma_i \le t \le \mu_i + D_i \\ 0 & \text{otherwise} \end{cases} \tag{24}$$

The raw shape is:

$$S_7(t) = \max\left(\sum_{i=1}^{2} G_i(t) + b,\ b\right) \tag{25}$$

Independent: $A_1 = 0.07$, $\mu_1 = 0.30$, $\sigma_1 = 0.08$, $D_1 = 0.25$; $A_2 = 0.09$, $\mu_2 = 0.80$, $\sigma_2 = 0.15$, $D_2 = 0.10$; $b = 0.042$. Place-dependent: $A_1 = 1.67$, $A_2 = 2.14$, same $\mu, \sigma, D$, $b = 1.0$.

## Type 8 — Post-CS Piecewise Decay

Independent: $b = 0.065$, with fixed absolute steps 0.03, 0.04, 0.025. Place-dependent: $b = 1.0$, with steps expressed as $0.46\,b$, $0.62\,b$, $0.38\,b$.

$$S_8(t) = \begin{cases} b & t \le 0.25 \\ s_1 & 0.25 < t \le 0.252 \\ \dfrac{s_2 - b}{0.498}(t - 0.252) + b & 0.252 < t < 0.75 \\ \dfrac{s_3 - b}{0.75}(t - 0.75) + b & 0.75 \le t \le 1.5 \\ b & t > 1.5 \end{cases} \tag{26}$$

where $(s_1, s_2, s_3) = (0.03, 0.04, 0.025)$ for the independent model, and $(s_1, s_2, s_3) = (0.46b,\ 0.62b,\ 0.38b)$ for the place-dependent model.

## Arousal-Mediated Model

*The task response is selected from one of 4 functions based on whether the rat is running and whether the cell is in a place field. Each of these functions depends solely on the time since CS onset and the neuron's own live place rate.*

There is no balance parameter and no place/task category split — every neuron is a place cell, and the task response (if TEBC-responsive) is added directly on top:

$$r_i(t) = \tilde{r}_i^{\text{place}}(t) + r_i^{\text{task}}(t) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}\big(-\tfrac{0.02}{30},\ \tfrac{0.02}{30}\big) \tag{27}$$

**Velocity modulation.** Uses the same cubic polynomial $p(s)$ as §2, with a different slow-speed floor. A randomly chosen subset of neurons, of size $(1 - \texttt{percent_place_cells}) \times N$, is additionally forced to a near-zero rate whenever the animal is running — *but not* when it is stopped, so this subset only reads as "non-place" during locomotion:

$$\tilde{r}_i^{\text{place}} = \begin{cases} \frac{0.2}{30}\, r_i^{\text{PC}} & v < 0.02 \\ \dfrac{p(100v)}{30}\, r_i^{\text{PC}} & v \geq 0.02,\ i \notin \mathcal{Z} \\ \frac{0.02}{30} & v \geq 0.02,\ i \in \mathcal{Z} \end{cases} \tag{28}$$

where $\mathcal{Z}$ is the randomly sampled zeroed-out index set (resampled once per simulation, not per timestep).

**Task firing rate.** The response function is selected by two binary conditions evaluated each timestep: whether the neuron is inside its place field, and whether the animal is running. Place-field membership is assessed from the *raw* (pre-velocity-modulation) place rate $r_i^{\text{PC}}$:

$$\text{in_field}_i = \big[r_i^{\text{PC}} \geq 0.2\big], \qquad \text{running} = \big[v > 0.02\big] \tag{29}$$

$$r_i^{\text{task}}(t) = \begin{cases} 0 & \text{neuron } i \text{ not TEBC-responsive} \\ R_1\big(t,\, b_i\big) & \text{in field,} \quad \text{running} \\ R_2\big(t,\, b_i\big) & \text{out of field,} \quad \text{running} \\ R_3\big(t,\, b_i\big) & \text{in field,} \quad \text{stopped} \\ R_4\big(t,\, b_i\big) & \text{out of field,} \quad \text{stopped} \end{cases} \tag{30}$$

where $b_i = \tilde{r}_i^{\text{place}}(t)$ is the velocity-modulated place rate used as the baseline for all magnitude expressions.

**Response functions.** $R_1$ — **In field, running.** Elevated through the CS and trace interval (the hippocampal "bridging" signature expected of a trace-conditioned cell), then peaks more strongly during the US pulse itself before returning to baseline:

$$R_1(t, b) = \begin{cases} 1.3\, b & t < t_{US} \\ 1.5\, b & t_{US} \leq t < t_{US} + d_{US} \\ b & t \geq t_{US} + d_{US} \end{cases} \tag{31}$$

$R_2$ — **Out of field, running:** a slight, sustained increase through the CS, trace interval, and US pulse:

$$R_2(t, b) = \begin{cases} 1.1\, b & t < t_{US} + d_{US} \\ b & t \geq t_{US} + d_{US} \end{cases} \tag{32}$$

$R_3$ — **In field, stopped:**

$$R_3(t,b) = \begin{cases} 1.5\,b & t < t_{CS} \\ b\left(1 + 0.5\,\dfrac{t_{US} - t}{t_{US} - t_{CS}}\right) & t_{CS} \le t < t_{US} \\ 3\,b & t_{US} \le t < t_{US} + d_{US} \\ b & t \ge t_{US} + d_{US} \end{cases} \tag{33}$$

$R_4$ — **Out of field, stopped:**

$$R_4(t,b) = b + \varepsilon_4, \qquad \varepsilon_4 \sim \text{Uniform}(-0.2\,b,\ 0.2\,b) \tag{34}$$